100 Days of Linux - Day 023

Title: Searching Text Inside Files with grep

Today's Goal
Learn how to search for words and patterns inside text files using the **grep** command.

Why This Matters
Developers, data engineers, and system administrators use grep every day to search logs, configuration files, SQL scripts, and source code.

Command of the Day: grep

Syntax: grep 'text' filename
Common Options:
grep -i 'text' file (ignore case)
grep -n 'text' file (show line numbers)
grep -r 'text' folder (search recursively)
Examples:
grep 'ERROR' app.log
grep -i 'python' notes.txt

Beginner Lesson
The grep command searches inside files instead of searching for file names. It prints only the lines that match your search. This makes it one of the most powerful Linux commands.

Practice Questions

1. Create a file named fruits.txt with at least 10 lines. Search for the word 'Apple' using grep.

2. Search for the word 'apple' without considering uppercase or lowercase letters.

3. Search for the word 'Apple' and display the matching line numbers.

4. Create a folder named logs with two text files. Put the word 'ERROR' in one file and use grep recursively to find it.

5. Display your current working directory after completing today's tasks.

Hints

Remember: grep searches inside files. Use -i for case-insensitive searches, -n for line numbers, and -r for folders.

Mini Challenge

Create a file named employees.txt with 15 records. Search for one employee name, then search for a department name using both grep and grep -n.

Common Mistakes

Using grep to search for file names. Use find or locate for files,

and grep for text inside files.

Did You Know?
The name 'grep' comes from the old editor command 'global regular expression print'.

Pro Tip
Combine grep with other commands later using pipes (|) to build powerful command-line workflows.

Answer Key (Instructor Only)
Q1: grep 'Apple' fruits.txt
Q2: grep -i 'apple' fruits.txt
Q3: grep -n 'Apple' fruits.txt
Q4: grep -r 'ERROR' logs/
Q5: pwd

Homework
Create three text files with different content and practice grep, grep -i, and grep -n on each one.

Next Day Preview
Tomorrow you'll learn Linux wildcards (*, ?, and []) to match multiple files and patterns.